HONOURS PROJECT

Noakhali Science and Technology University

Department of Applied Mathematics

According to B.Sc. Syllabus 2019-20

Project Tittle on

"Mathematical Statistics Approach to Evaluate the Effect of Volunteering on Higher Studies at Noakhali Science and Technology University"

Submitted By

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Session: 2019-20
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Dedicated

TO

My beloved parents

&

respected teachers

CERTIFICATION

This is to cerify that the project work entitled ***"Mathematical Statistics Approach to Evaluate the Effect of Volunteering on Higher Studies at Noakhali Science and Technology University",***is a research project work carried out by **Arman Hossen (MUH2006008M)**, the author under my supervision partial fulfillmentof the requirements for the degree of "Bachelor of Science" in Applied Mathematics, under the department of Applied Mathematics ,Noakhali Science and Technology University.

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Finally, I am personally thankful to my group members who provided expertise that greatly assisted on this project paper. Their contribution and support made my work a lot easier. In addition, of course, nothing could have come true without the support of my family for their constant support.

The Author

ABSTRACT

Research indicates that participation in "the effect of volunteer activities on higher studies of Noakhali Science and Technology University students " More specifically, studies have been conducted assessing the effects of specific extra-curricular activities on academic performance. The purpose of this study was to determine whether or not the activities in which students of higher education choose to participate have an effect on their academic performance. The study survey instrument which was distributed to the students of Noakhali Science and Technology University who are involved in volunteer organizations like "(BNCC,Rover Scout ,NSTUDS,NSTU Blood Brigade, Luminery, Cholo paltai foundation ,MUN)". Mainly We have tried to find out the relevant effect of spending time in volunteering organization with academic performance. Therefore ,it was concluded that such kind of extra-curricular activities affect academic performance and the effect depends on the specific activities in which the students is involved.

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CHAPTER ONE

RESEARCH PERSPECTIVE

1.1 Introduction

Volunteering has become an essential part of our education and social security sector for a better society. Millions of people worldwide participate in voluntary services to contribute their merit and energy to make a more skillful and happy nation. To improve their lives, volunteers and voluntary organizations take flexible and persuasive programs. In Bangladesh, thousands of students are involved in voluntary organizations and services for a better community. It is considered that Dhaka University is the 'cultural hub' of Bangladesh. For example, in every national fest, such as in "Pahela Boishakh" (The first day of the Bangla calendar), Hundreds of students participate in voluntary services. Hundreds of students expand their hands to the destitute people during the national crisis, such as heavy floods, cyclones, storms, earthquakes, and other problems. They provide mental and material support to the affected people. In addition, volunteers campaign on campus and across the country to raise social awareness against rape and other social injustice against girls and women. In the last couple of years, hundreds of students have been sent to peripheral areas of Bangladesh to protect against injustice like dowry, and torture. In addition, students talked with people about the destructive impacts of drug addiction, early marriage, and poverty on individual and societal, and national levels. Volunteering can improve students' academic and personal skill development and sense of civic responsibility and explore career choices and employability after graduation (Holdsworth & Quinn, 2010). Therefore, voluntary service is urgent for the students and the nation. To make volunteering a routine and habitual work, we must come forward to encourage university students.

Now-a-days a student is not only judge by his academic performance. He is judged by his result as well as his efficiency in presentation ability and innovative mind. And one of the ways acquired these types of attributes through working on volunteer organizations. For this reason,

We choose the research topic on "the effect of volunteer activities on academic performance of Noakhali Science and Technology University students"

1.2 Objectives of Study

Every research is conducted on the base of some specific objectives. The objectives of the study are:

a. To investigate academic results of students who are working in volunteer organizations.
b. To know whether working in volunteer organization cause any bad effect on their academic result.
c. At what sex people mainly involve in this work?
d. To investigate whether it helps to improve student's quality like their communication skill, leadership and organization management ability etc.

1.3 The history of volunteer organizations and their activities:

The development of volunteer organization considers as extra-curriculum activities was slow in the beginning with many seeing it simply as a fad that would pass and quickly fade out of style (millard,1930, p. xi). One of the early philosophies behind extra curriculum activities that they should wherever at all possible, "Grow out of curricular activities and return to curriculum activities to enrich them"(millard, 1930, p.12). Eventually people including educators began to see the benefits of extra-curricular activities but it took a while to ensure themselves to them. In fact before 1900, educators were sceptical of participation in volunteering organization believing "school should focus solely on narrowly defined academic outcomes". Non-academic activities moreover including helping organization were viewed as being primarily recreational and therefore were detrimental to academic achievement and consequently were discouraged" (marsh & kleitman, 2002, para. 5). Dream and bear, early experts on extra-curricular activities said "Extra-curricular activities supplement and extend those contacts and experiences found in the more formal part of the program of the school day "(millard 1930, p 16). Wet was not until recently that " Educational practitioners and researchers have taken a more positive perspective arguing that extracurricular activities may have positive effects on life skills and may also benefit academic accomplishments (marsh & kleitman, 2002, para 5). It is obvious that extra-curricular activities have an impact on academic performance and education ever since their inception. The question is how are extra-curricular activities affecting academic performance today? Activities, such as watching television & use social media. Some literature on leisure studies has "Suggested that formal and in formal activity settings have different influence on motivation and feelings of competence", two factors which influence academic performance (Guest & Schneider,

2003, para. 8). One study found "That more time in leisure activities was related to poorer academic grades, poorer work habits, and poorer emotional adjustments" while more time in "structured groups and less time watching tv & social media were associated with higher test scores and school grades" (marsh & kleitman 2002, para 15). Guest and Schneider (2003) in their study found that "The type of participation or activity undertaken influences development outcomes (para .8). This involves the "what" factor and is the concert of this research project. There have been many studies conducted on the influence that extra-curricular activities have on academic performance.

CHAPTER TWO

METHODOLOGY

2.1 Introduction

This chapter presents the methodology of the study which consists of the followings: Research design, population, sample and sampling techniques, developing instrument, data collecting procedures, data input and analysis, data triangulation and ethical consideration.

2.2 Place and Time of the Data Collection

For this research, data were collected from different students of Noakhali science and Technology university who works in volunteer organizations from July to August of 2024 .

2.3 Preparation of the Questionnaire:

The important part of the sample survey is the construction of the questionnaire (to be filled by respondent). Require skill, special technique as well as familiarity with the subject matter under study. A questionnaire is an important element for collecting data in any project work. To make a good sufficient questionnaire we need special skill and experience. A questionnaire must be understood to the respondents, brief, collaborative and not offending.

The question should be as many as it is not the matter of hesitation to the respondents. After preparing the questionnaire We gave it to our supervisor for necessary correction. My questionnaire contains only 34 questions and some questions are categorical type. All questions are very simple. My question was written in Bengali Language.

2.4 Procedure of data collection:

The data should be collected keeping in view the objects of the survey. The editing of completed questionnaire helped in amending and recording errors or eliminating data that are obviously erroneous and inconsistent. All kinds of mistakes have been corrected which were found in our questionnaires and all the answers have been observed carefully. As a result, We feel there is no incomplete and irrelevant information. The tendency should not be to collect too examined and analyzed.

The process of data collection depends upon the kind of research generally

1. Mailing process
2. Interview process

The interview process has two parts:

1. Questionnaire process
2. Focus group discussion

We have selected questionnaire process for our study.

2.5 Sampling method:

We study the effect of working in volunteering organization on the student of Noakhali science and Technology university . For our research project our population is the students of Noakhali science and Technology university . It is not possible to carry out information from all the students of this institutions because of time constancy and costs. At beginning of the study, We decide that We only collect those data that are spent time in volunteering organization. All students of Noakhali science and Technology university were considered as the population in this study. Sample size determination formula provided 185 students were the aquacade sample size for the present study. However, we selected 200 samples for the study. Two-stage random sampling was used for selecting sample from the population. In the first stage, six faculties were selected in Noakhali science and Technology university. In the second stage, students who works as volunteer were selected randomly from each selected faculty. Before collecting data, the objective of our study was discussed with all selected students, took written consent. Data was collected using self-developed pre-tested structured questionnaire. Appropriate statistical tools/models were used for analysis data. A value of $p<0.05$ was considered as statistically significant in the analysis. All statistical analyses were performed using SPSS (IBM Version 30).

2.6 Data editing:

After collecting the necessary data, at first We edit those data orderly as it can be easily analyzed in computer for getting expected result of the study. Without editing the data, it is very difficult to analyze properly. So data editing is very important part in any research. All kinds of mistakes have been corrected which were found in our questionnaire and all the answers have been observed carefully. As a result, We think it had no incomplete information.

2.7 Selection of computer program and data entry:

The easiest procedure of analyzing the data is to use computer program. At present nobody thinks of analyzing the data without a suitable computer program. So We had to select a suitable computer program for data entry and analyze. We had selected SPSS for windows Version 20.0.

2.8 Analysis of data:

The analysis of data is an important part of any Research Project. At first We coded all the variables of the questionnaire according to the predetermined code plan and enter the data into the computer. Then We made different types of table, Bar chart, Histogram, pie chart, cross table. We also prefer some graphs to make a decision.

2.9 Data processing:

The procedure of data processing consists of editing, coding, and tabulation has done by the researcher. The analysis has done with the help of computer. A brief description about data processing is as follows.

1) Coding:- After editing the collected data we coded the possible answer on questionnaire. Coding refers to the process of assigning numbers or symbols to answer so that responses can be put into a limited number of categories and classes. It makes the process easier and faster. The coded data are checked with the original data.
2) Tabulation:- Tabulation is a process of summarizing raw data.

2.10 Problem in collecting data:

During the period of data collection of our project, We have faced some problems. The problems that We faced are as follows:-

a) Many respondents hesitated to give answer of some belonging to the questionnaire for educational level.
b) Respondent may give false answer.
c) Since most of the respondents did not say their family income and expenditure correctly.
d) Some of the respondents can not say his/her parent's educational level.

2.11 Limitation of survey:

Every work or survey which are done has some limitation, some are given below:

a) Proper care should be taken in the planning and execution of the survey otherwise the result obtained might be incorrected and misdealing.
b) Survey requires the service of trained and qualified person and sophisticated equipement for it's planning, execution and analysis. In the absence of these the results of the survey will not trustworthy.

Beside these, we need enough time, money and adequate skilled personal to conduct a survey.
As We are student of Applied Mathematics Department of Noakhali science and Technology university to give enough time and is hardly for us. Also we are not as skilled as require.

2.12 Methodology

After processing data, at first we extract some frequency distribution from the data. We also construct some bar diagram and some pie chart.

Frequency distribution: A frequency distribution is a tabular summary of data where observations are divided into different non overlapping classes or categories and frequency of each class or category is arranged accordingly. Sometimes, qualitative data are shown as relative frequency distribution or percent frequency distribution.A percent frequency distribution is tabular summary of data showing the percent frequency for each class.

Bar diagram: Bar diagram is the most popular diagrammatical presentation of qualitative data. By simple bar diagram only one qualitative data can be exhibited. First a frequency distribution table for qualitative is constructed. Then a bar of a fixed width above each category or class is drawn.The main limitation of bar diagram is that only one qualitative variable can be displayed through these. More than one qualitative variables can be exhibited by multiple or component bar diagram.

Pie Chart: A pie chart is used to show how a whole is divided among several categories. The chart is in the form of circle and is also called a Pie, because the chart looks like a pie and the components resemble slices cut from it. The size of the slice represents the proportion of the component out of the total.

Contingency Analysis: Contingency analysis is the approach we will use to test whether or not there is an association between two categorical variables. As with the goodness-of-fit test we can use a chi-square test statistic to test our null hypothesis.

In previous chapters, the background of the study and the method of data collection and computerization are discussed. Generally it is important to know the characteristic or nature of the data before performing any statistical analysis. In order to know the nature and characteristics of the data, frequency distribution and graphical representation could be very useful.

Frequency Analysis: At first We would like to present frequency distribution and relevant graphs to have a quick look of our data.

Faculty Status

Table-2.13.1: Frequency distribution of Faculty:

Faculty	Frequency	Percent
Science	100	50.0
Education	39	19.5
Business	17	8.5
Social Science	28	14
Engineering	10	5.0
Law	6	3.0
Total	200	100.0

Table-2.13.1 shows that most of the respondents 50% are from Science faculty, then 19.50% from Education faculty, 8.5% from Business faculty, 14% from Social Science faculty, 5% from Engineering faculty and 3% from Law faculty.

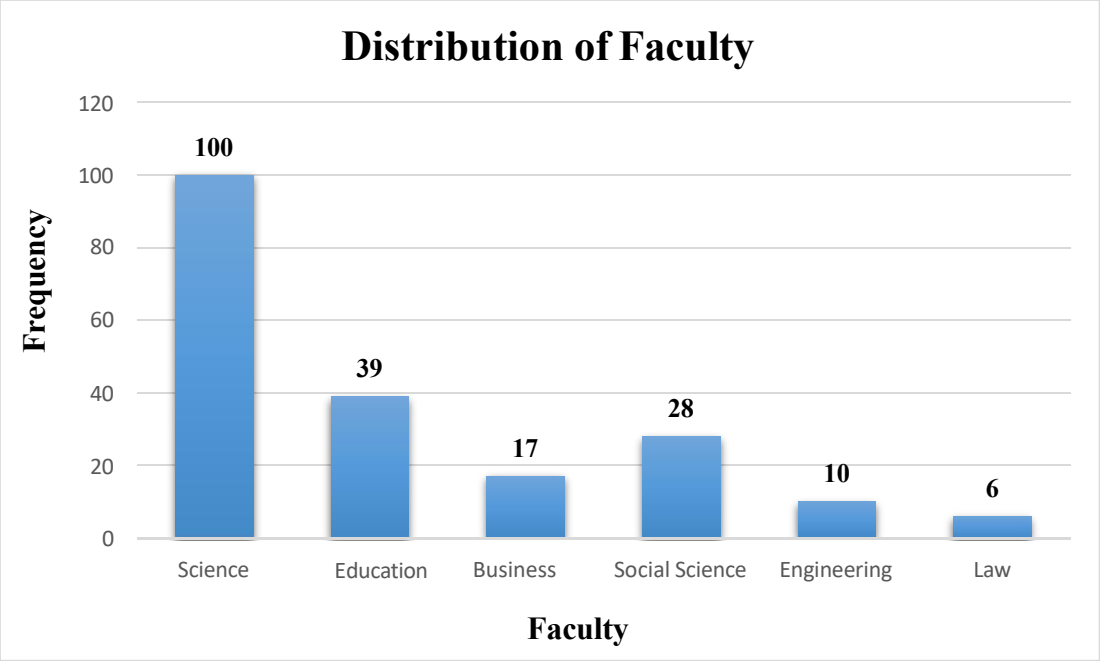

Figure-2.13.1: Bar-diagram of Faculty

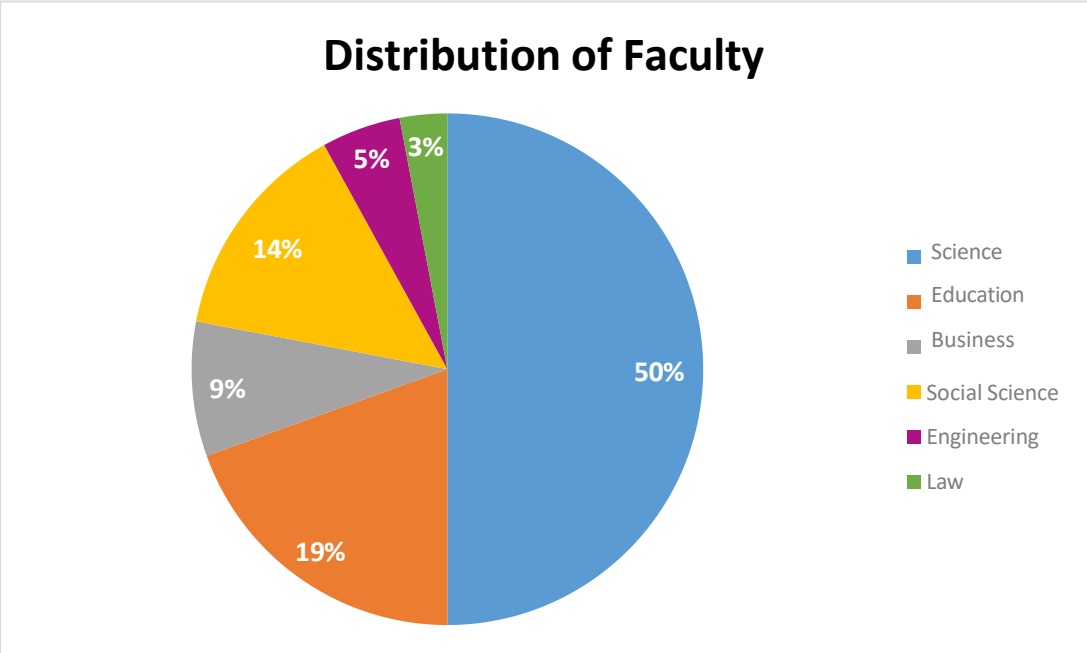

Figure-2.13.1: Pie-chart of Faculty

Academic Year

Table-2.14.2: Frequency distribution of Academic Year:

Academic Year	Frequency	Percent
1st Year	25	12.5
2nd Year	37	18.5
3rd Year	53	26.5
4th Year	69	34.5
Master's	16	8.0
Total	200	100.0

Table-2.14.2 shows that most of the respondents 34.5% are from 4th year, then 12.5% are from 1st year, 18.5% are from 2nd year , 26.5% respondents are from 3rd year and 8% respondents are from master's.

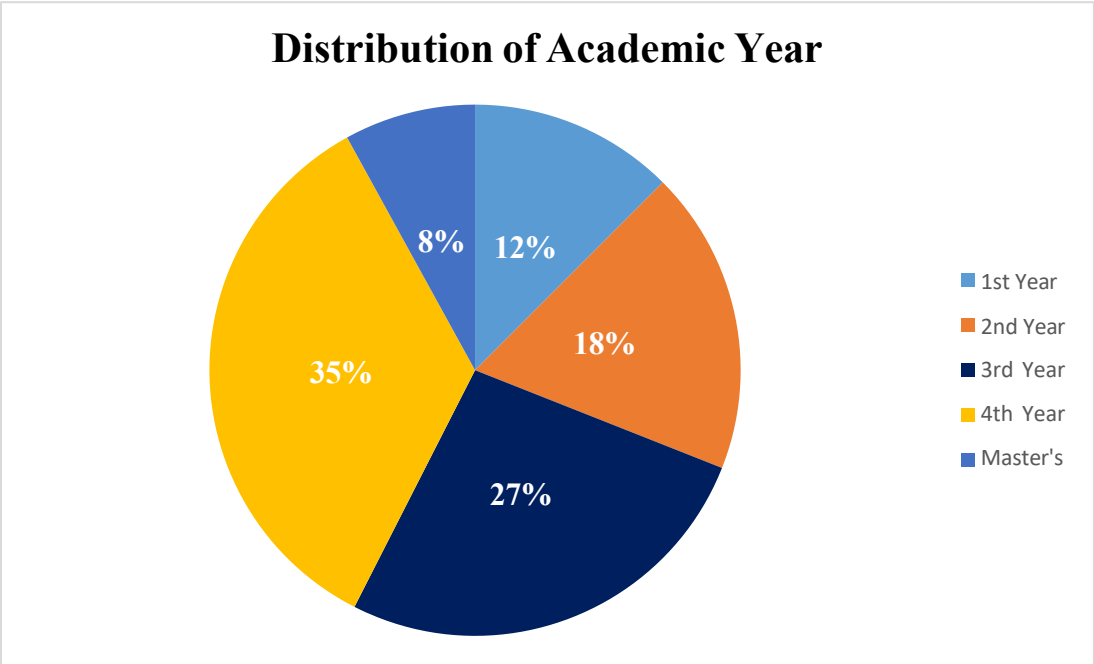

Figure-2.14.2: Pie-chart of Academic Year

Age Status

Table-2.15.3: Frequency distribution of Respondent's Age:

Respondent's Age	Frequency	Percent
<=20	24	12.0
21-23	119	59.5
23+	57	28.5
Total	200	100

Table-2.15.3 shows that most of the respondents 59.5% are between the interval 21-23 years old, then 12% are below and 28.5% are above 23 years old.

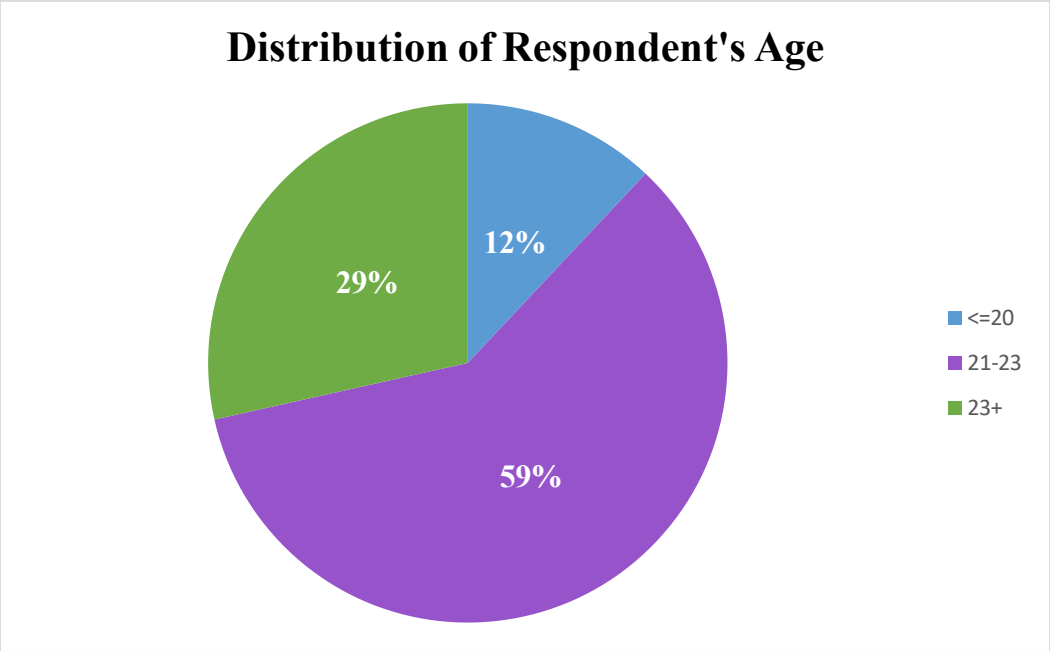

Figure-2.15.3: Pie-chart of respondent's age

Religion Status

Table-2.16.4: Frequency distribution of Religion:

Religion	Frequency	Percent
Muslim	171	85.5
Others	29	14.5
Total	200	100.0

Table-2.16.4 shows that 85.5% respondents are Muslim and 14.5% are from others religion.

Gender Status

Table-2.17.5: Frequency distribution of Gender status:

Sex	Frequency	Percent
Male	178	89.0
Female	22	11.0
Total	200	100.0

Table-2.17.5 shows that 89% respondents are male and 11% are female.

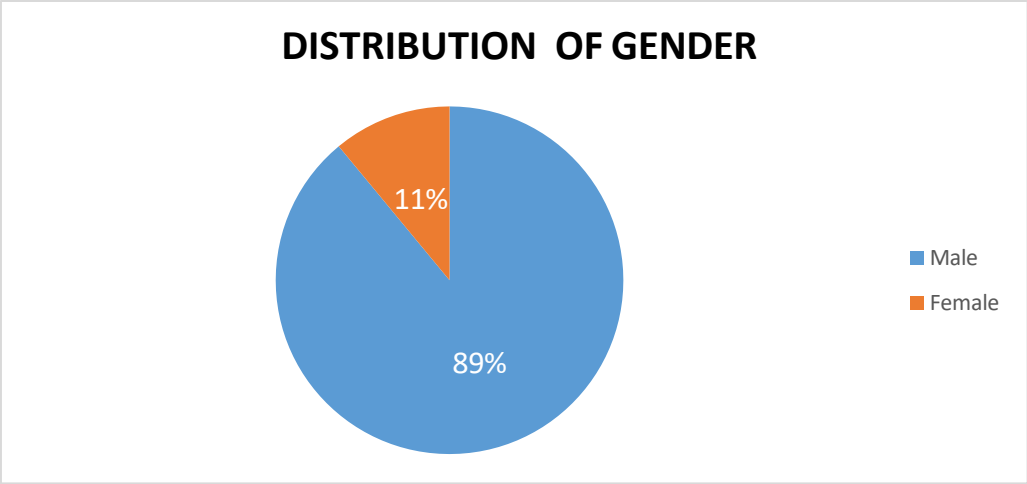

Figure-2.17.5: Pie-chart of gender status

Respondent's Division

Table-2.18.6: Frequency distribution of Respondent's Division:

Respondent's division	Frequency	Percent
DHAKA	23	11.5
RAJSHAHI	46	23
CHITTAGONG	30	15
SYLHET	01	05
RANGPUR	60	30
KHULNA	31	15.5
BARISAL	02	01
MYMENSINGH	07	3.5
Total	200	100

Table-2.18.6 shows that most of the respondents 30% are from Rangpur, then 11.5% are from Dhaka, 23% are from Rajshahi, 15% are from Chittagong, 5% are from Sylhet, 15.5% are from Khulna, 1% are from Barisal and 3.5% are from Mymensingh.

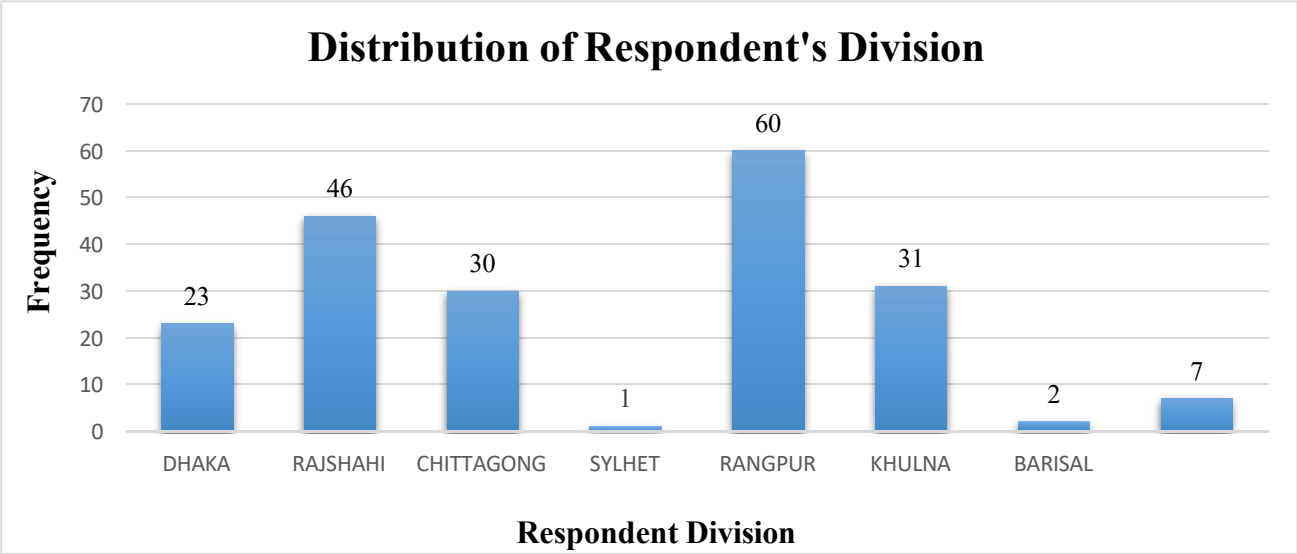

Figure-2.18.6: Bar-chart of Respondent's Division

Father's Education

Table-2.19.7: Frequency distribution of Father's Education:

Father's Education	Frequency	Percent
Primary	44	22.0
Secondary	42	21.0
Higher Education	83	41.5
Illiterate	31	15.5
Total	200	100.0

Table-2.19.7 shows that 22% respondent's fathers are primary educated, then 21% are secondary educated, 41.5% are higher educated and 15.5% are illiterate.

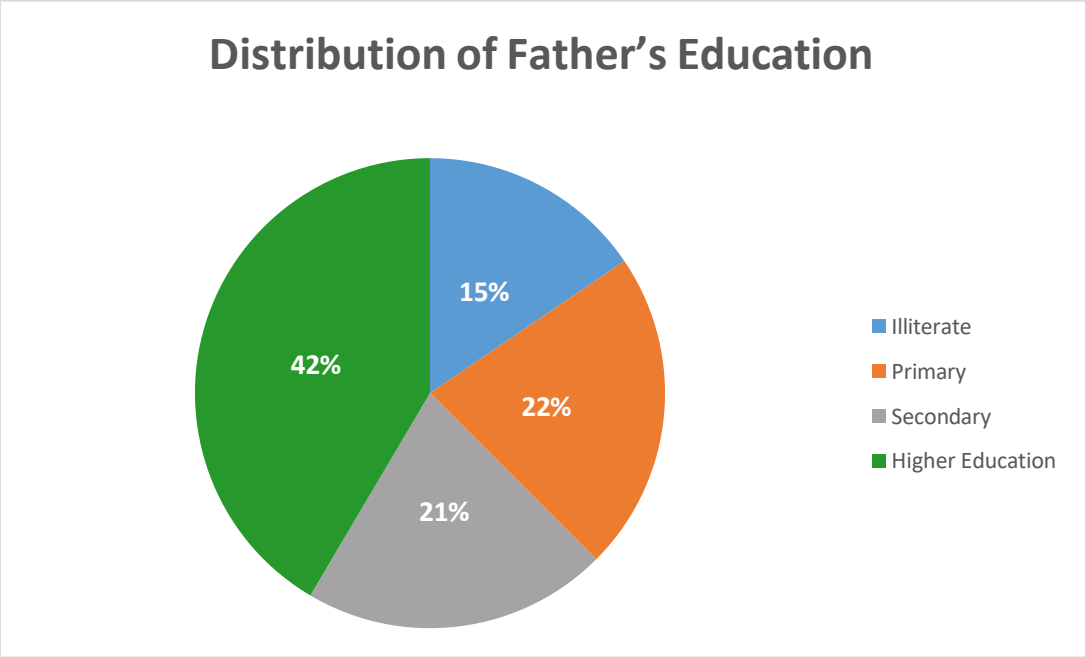

Figure-2.19.7: Pie-chart of Father's education

Father's Occupation

Table-2.20.8: Frequency distribution of Father's Occupation:

Father's Occupation	Frequency	Percent
Govt. Service	32	16.0
Business	55	27.5
Farmer	69	34.5
Teacher	18	9.0
Others	26	13.0
Total	200	100.0

Table-2.20.8 shows that most of the respondents fathers 34.5% are farmers, then 16% are government job holders, 27.5% are businessmen, 9% are teachers and 13% are doing other jobs.

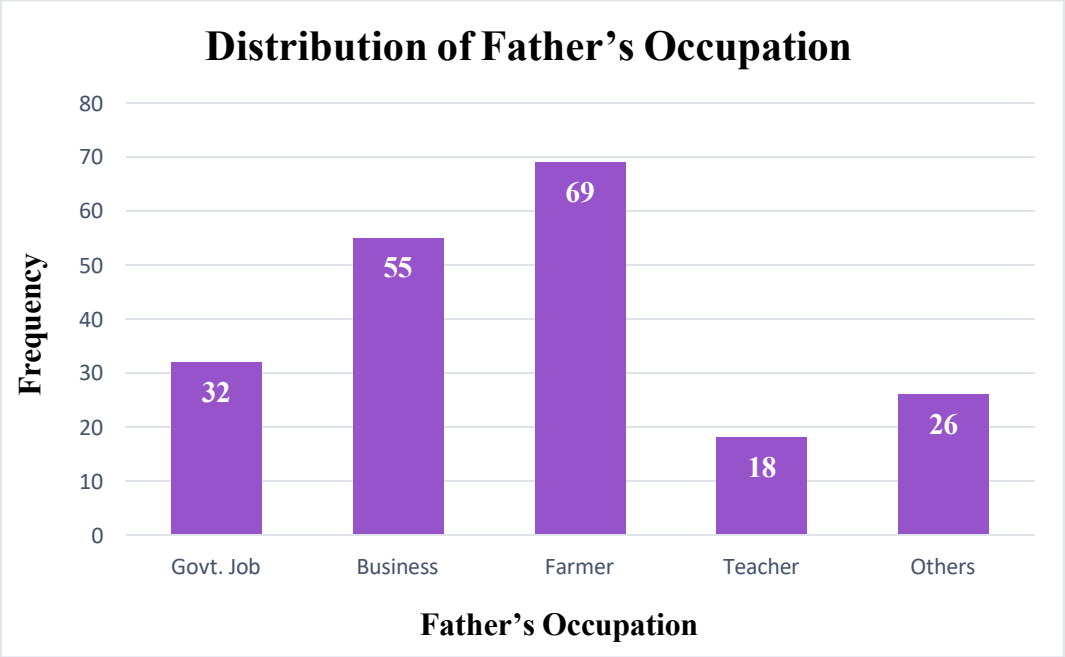

Figure-2.20.8: Pie-chart of Father's occupation.

Mother's Education

Table-2.21.9: Frequency distribution of Mother's Education:

Mother's Education	Frequency	Percent
Primary	63	31.5
Secondary	53	26.5
Higher Education	51	25.5
Illiterate	33	16.5
Total	200	100.0

Table-2.21.9 shows that most of the respondents mothers 31.5% are primary educated, then 26.5% are secondary educated, 25.5% are higher educated and 16.5% are illiterate.

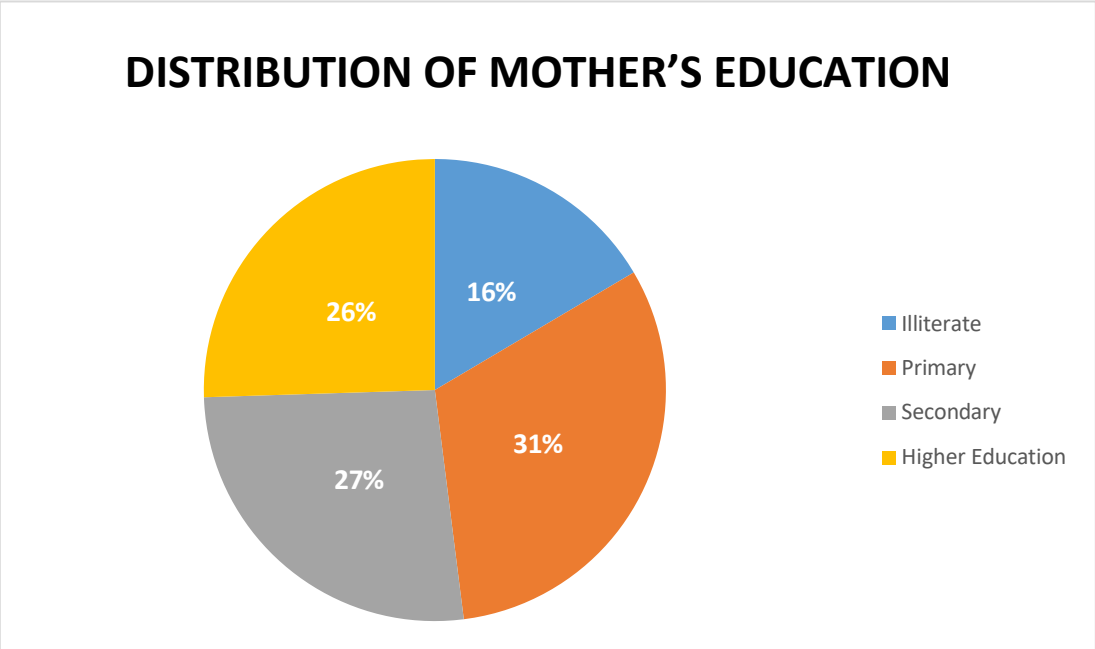

Figure-2.21.9: Pie-chart of Mother's education

Mother's Occupation

Table-2.22.10: Frequency distribution of Mother's Occupation:

Mother's occupation	Frequency	Percent
Teacher	11	5.5
Housewife	173	86.5
Others	16	8.0
Total	200	100.0

Table-2.22.10 shows that most of the respondents mothers 86.5% are housewives, then 5.5% are teachers and 8% are doing other jobs.

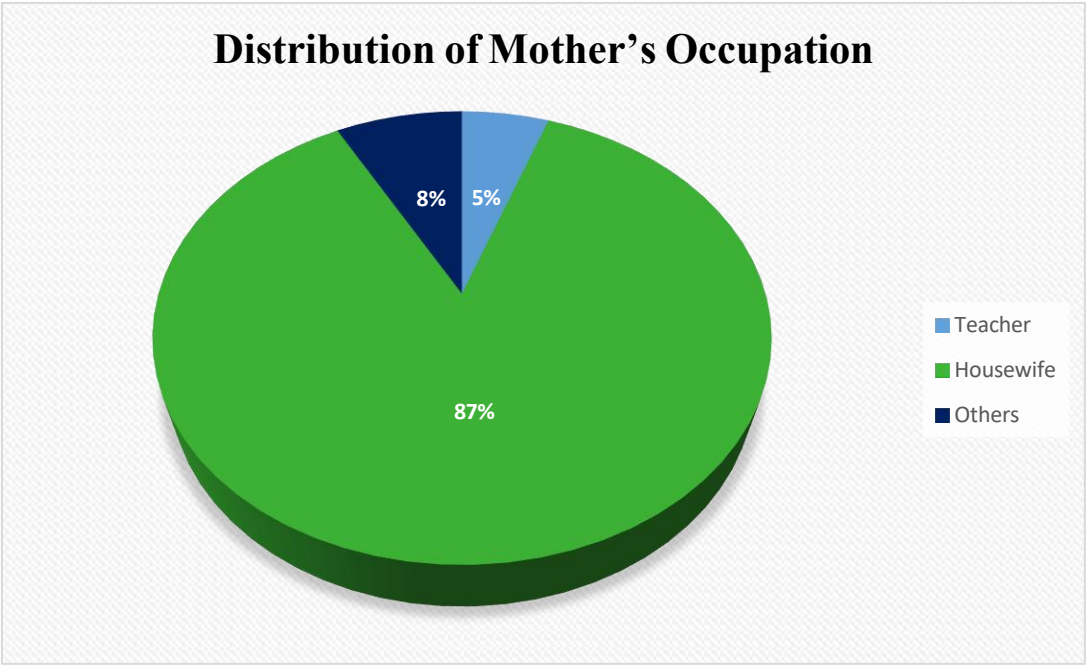

Figure-2.22.10: Pie-chart of Mother's occupation

Family Income

Table-2.23.11: Frequency distribution of family income:

Family Wencome (Monthly)	Frequency	Percent
>=10000	45	22.5
10001 - 20000	73	36.5
20001 - 30000	40	20.0
30001 - 40000	19	9.5
40000 +	23	11.5
Total	200	100.0

Table-2.23.11 shows that the most of the respondent's (36.5%) family income is between the interval 10001-20000 taka.

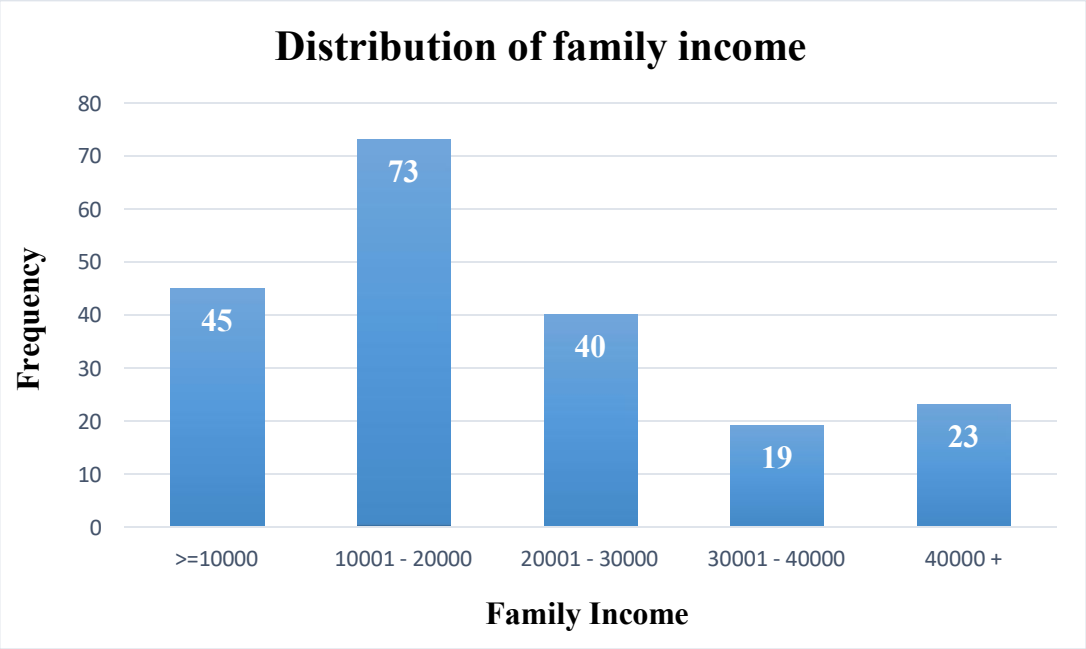

Figure-2.23.11: Pie-chart of Family Income

Family Type

Table-2.24.12: Frequency distribution of Family Type:

Family Type	Frequency	Percent
Nuclear	171	85.5
Joint	29	14.5
Total	200	100.0

Table-2.24.12 shows that 85.5% respondents are from nuclear family and 14.5% are from joint family.

Types of Volunteer Organizations

Table-2.25.13: Frequency distribution of volunteer Organizations:

Organization Name	Frequency	Percent
Cholo Paltai Foundation	44	22.0
Rover Scout	54	27.0
BNCC	12	6.0
NSTUDS	20	10.0
NSTU Blood Brigade	9	4.5
Luminery	22	11.0
Others	39	19.5
Total	200	100.0

Table-2.25.13 shows that most of the respondents (27%) are working in Rover Scout, then (22%) are in Cholo Paltai Foundation , (6%) are in BNCC, (10%) are in NSTUDS, (4.5%) are in NSTU Blood Brigade, (11%) are in Luminery and (19.5%) are in other volunteer organizations.

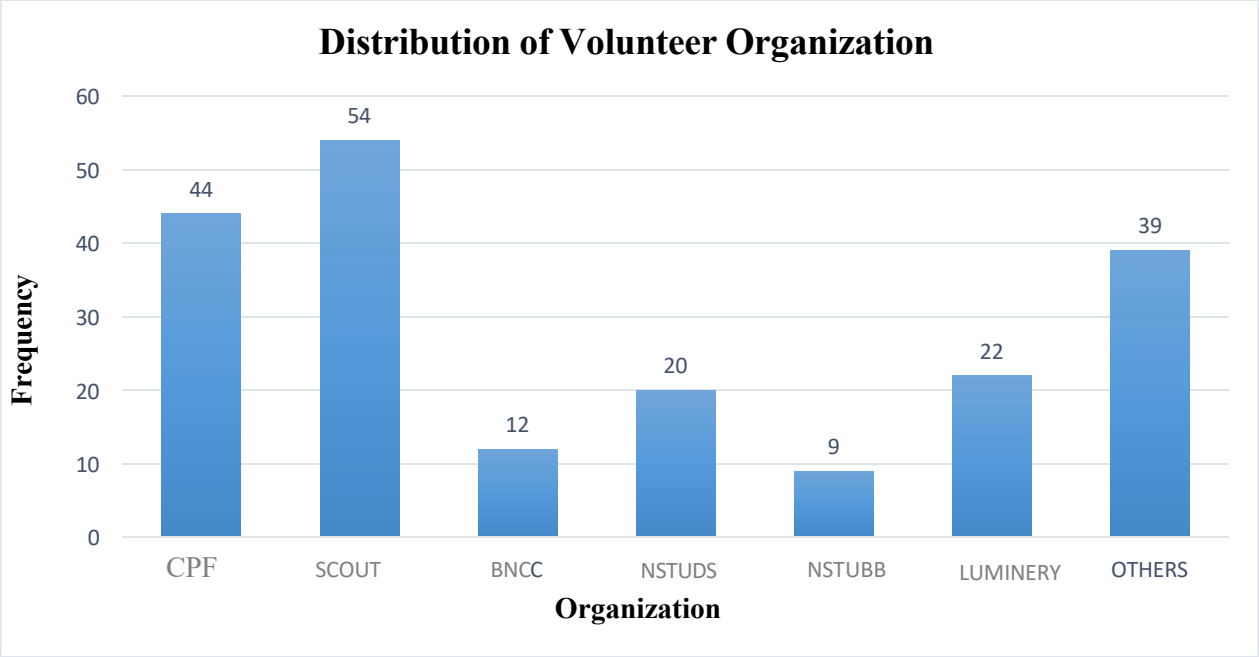

Figure-2.25.13: Bar-diagram of Organization name

Involvement in Multiple Organizations

Table-2.26.14: Frequency distribution of involvement in Multiple Organizations:

Multiple Organization	Frequency	Percent
No	117	58.5
Yes	83	41.5
Total	200	100.0

Table-2.26.14 shows that 41.5% respondents are involved in multiple organizations and 58.5% are in individual organization.

Spending time in Volunteer Services

Table-2.27.15: Frequency distribution of spending time in volunteer services:

Spending time	Frequency	Percent
LESS THAN 1 HOUR	40	20.0
1 HOUR	107	53.5
MORE THAN 1 HOUR	53	26.5
Total	200	100.0

Table-2.27.15 shows that most of the respondents (53.5%) spend 1 hour, then (20%) spend less than 1 hour and (26.5%) spend more than 1 hour in their volunteer organizations.

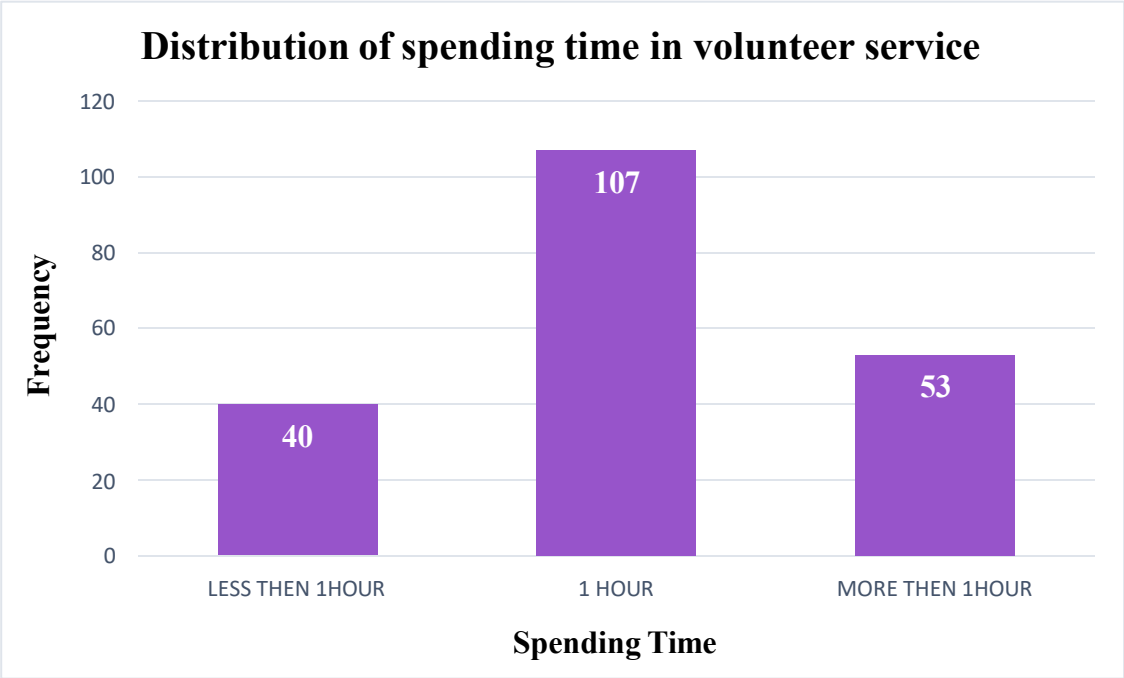

Figure-2.27.15: Bar-chart of spending time.

Participation in Volunteer Organization's Activities

Table-2.28.16: Frequency distribution of Participation in Volunteer Organization's Activities:

Participation in Organization's Activities	Frequency	Percent
SOCIAL WELFARE	10	05
BLOOD DONATION	76	38
CHILD EDUCATION	30	15
WINTER CLOTH DISTRIBUTION	10	05
ADMISSION DUTY	16	08
OTHERS	58	29
Total	200	100

Table-2.28.16 shows that most of the respondents (38%) participate in blood donation in their volunteer organization.

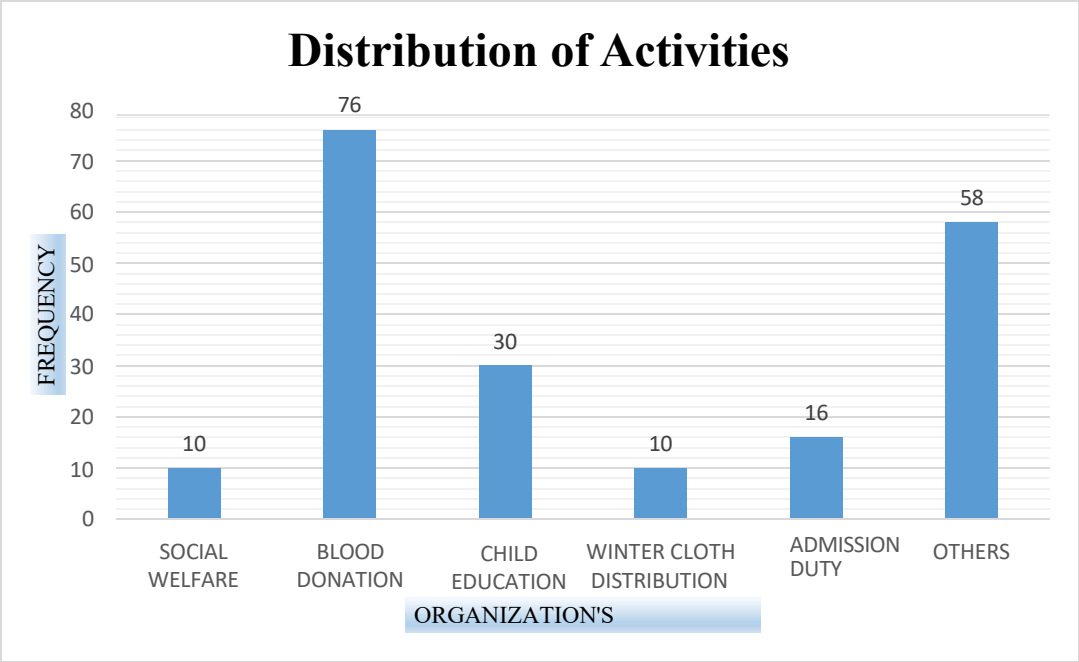

Figure-2.28.16: Bar-diagram of organization's activities

Having Bad Effect on Study

Table-2.29.17: Frequency distribution of Having Bad Effect on study:

Bad effect on Study	Frequency	Percent
No	169	84.5
Yes	31	15.5
Total	200	100.0

Table-2.29.17 shows that 84.5% respondents are agreed that working in volunteer organization does not have bad effect on their study and 15.5% are disagreed.

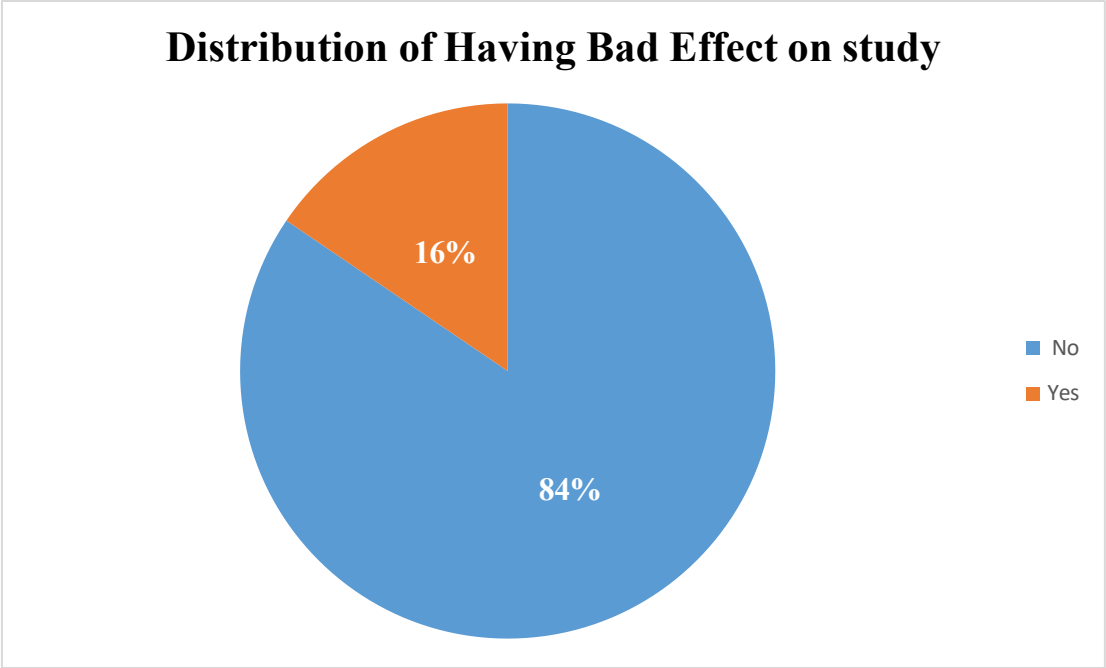

Figure-2.29.17: Pie-chart of bad effect on study

Activity Like the Most

Table-2.30.18: Frequency distribution of Activity like the Most:

Activity Like Most	Frequency	Percent
SOCIAL WELFARE	59	29.5
BLOOD DONATION	57	28.5
CHILD EDUCATION	26	13
WINTER CLOTH DISTRIBUTION	16	08
ADMISSION DUTY	31	15.5
OTHERS	11	5.5
Total	200	100

Table-2.30.18 shows that most of the respondents (29.5%) like to do social welfaring, then (28.5%) like blood donation, (13%) like child education, (8%) like winter cloth distribution, (15.5%) like Admission duty and (5.5%) like other activities in their volunteer organizations.

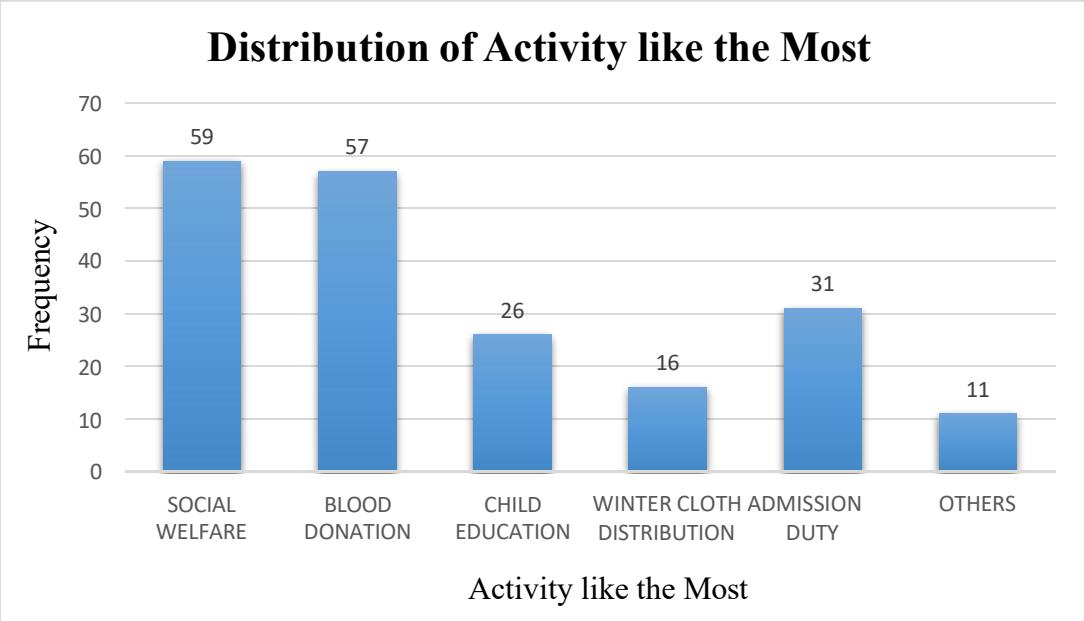

Figure-2.30.18: Bar-diagram of activity like the most

Reason of Volunteering

Table-2.31.19: Frequency distribution of Reason of Volunteering:

Reason of Volunteering	Frequency	Percent
PUBLIC WELFARE	50	25.0
OWN INTEREST	89	44.5
SOCIAL RESPONSIBILITY	25	12.5
OTHERS	36	18.0
Total	200	100.0

Table-2.31.19 shows that most of the respondent's (44.5%) volunteering reason is their own interest, then (25%) cause for public welfare, (12.5%) are for social responsibility and (18%) are for others interest.

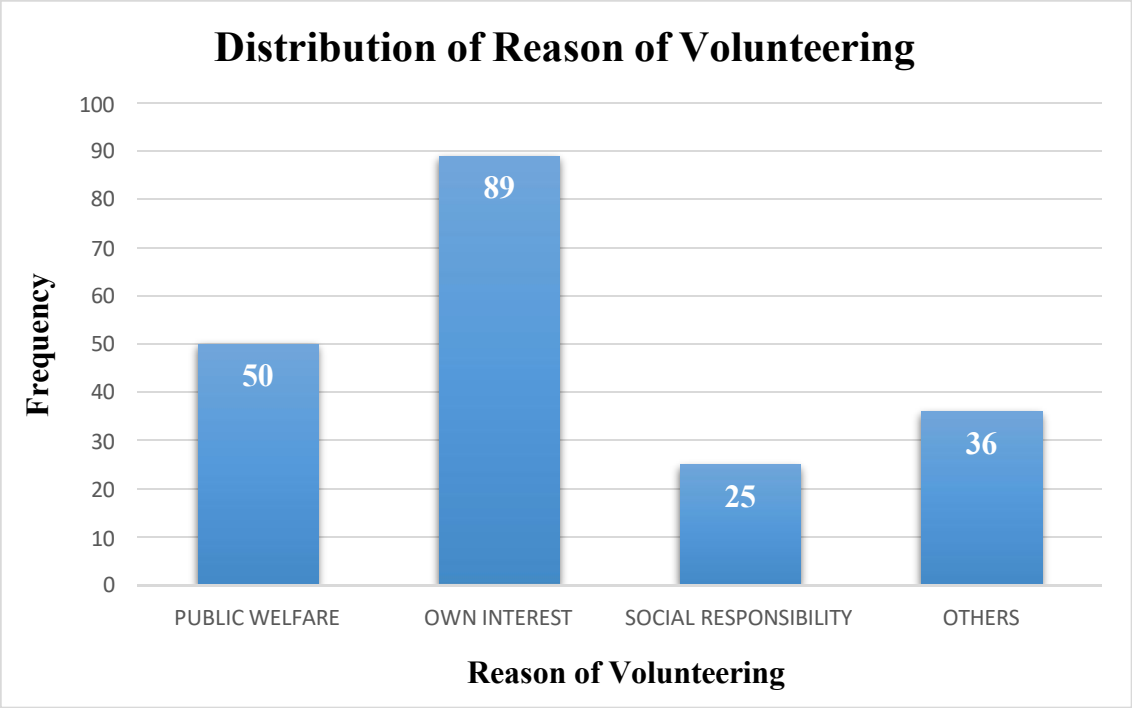

Figure-2.31.19: Bar-diagram of Reason of Volunteering

Having Progress

Table-2.32.20: Frequency distribution of having progress in life:

Progress	Frequency	Percent
No	42	21.0
Yes	158	79.0
Total	200	100.0

Table-2.32.20 shows that 79% respondents are agreed that they have made progress in life involving in volunteer organizations and 21% are disagreed.

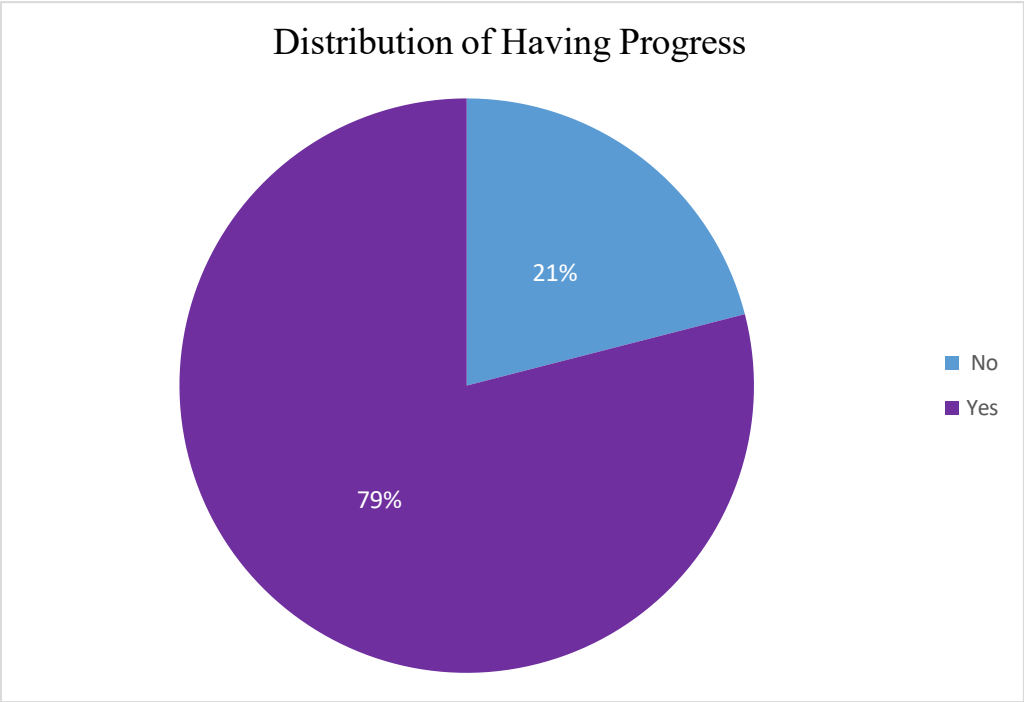

Figure-2.32.20: Pie-chart of progress

Progress Type

Table-2.33.21: Frequency distribution of Progress Type:

Progress Type	Frequency	Percentage
ORGANIZING & MANAGEMENT ABILITY	27	13.5
COMMUNICATIVE SKILL IMPROVEMENT	72	36.0
ACHIEVE LEADERSHIP QUALITY	23	11.5
NONE	41	20.5
OTHERS	37	18.5
Total	200	100

Table-2.33.21 shows that most of the respondents (36%) have developed their communication skill, then (13.5%) developed organizing & management ability, (11.5%) developed leadership quality, (18.5%) developed others quality and 20.5% are said that they have made no progress.

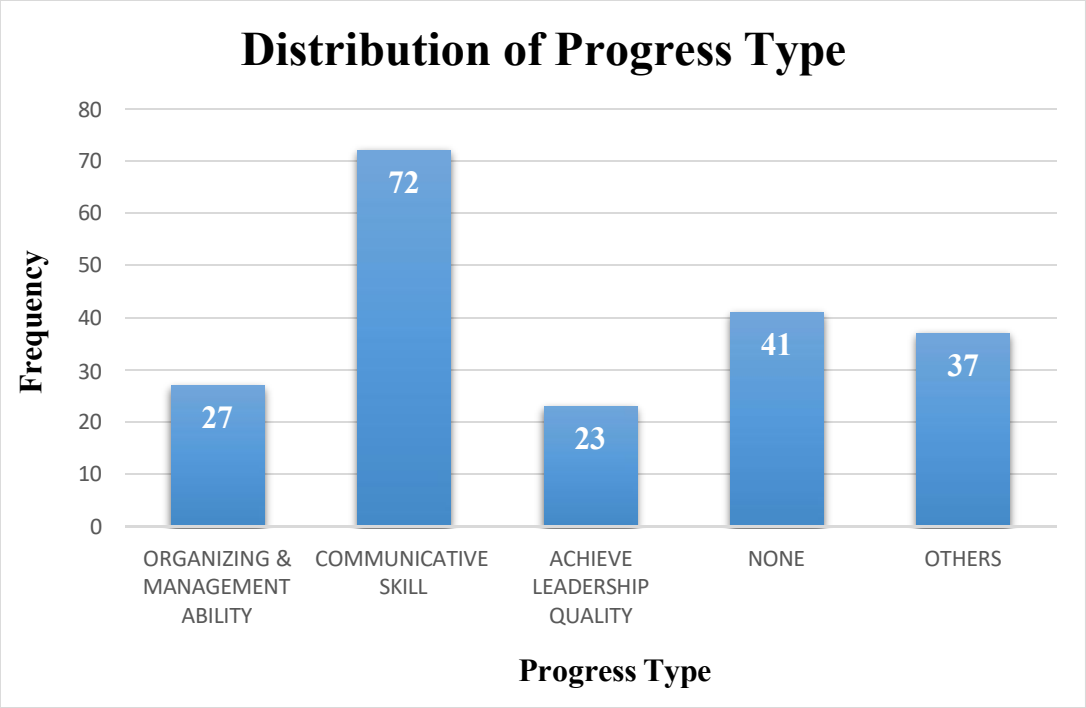

Figure-2.33.21: Bar-chart of Progress Type

Involvement before Higher Study

Table-2.34.22: Frequency distribution of involvement before higher study:

Involvement before higher study	Frequency	Percentage
No	141	70.5
Yes	59	29.5
Total	200	100.0

Table-2.34.22 shows that 29.5% respondents are saying that they were involved in volunteer organization before higher study and 70.5% were not.

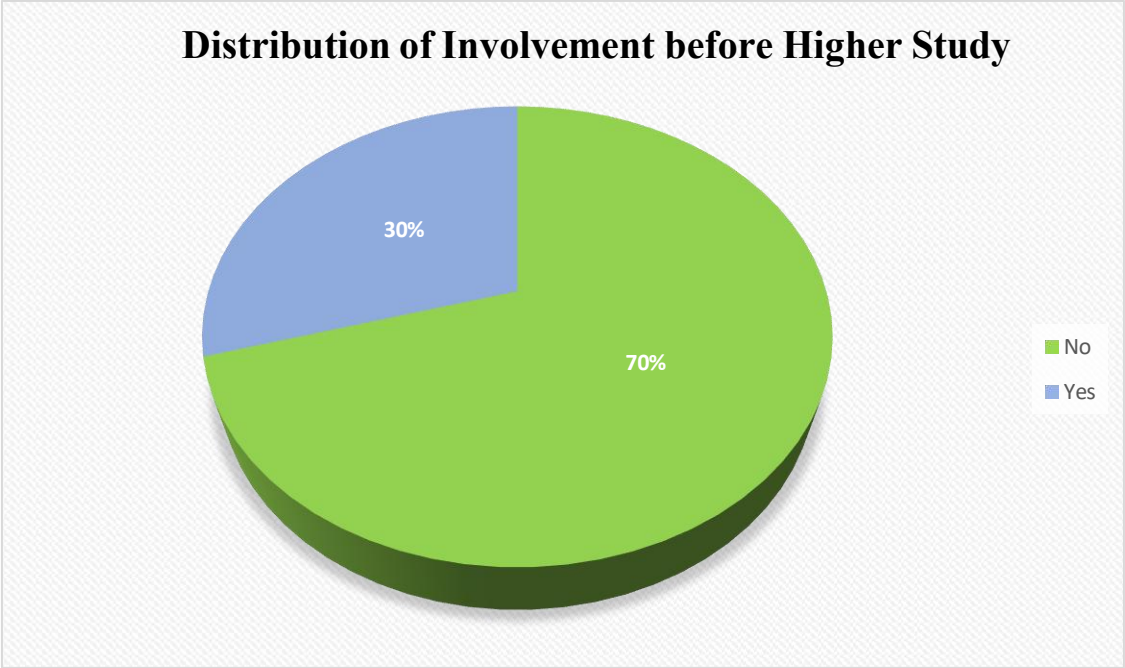

Figure-2.34.22: Pie-chart of involvement before higher study in volunteer organization

Future Plan

Table-2.35.23: Frequency distribution of Future Plan:

Future Plan	Frequency	Percent
GOVT. SERVICE	77	38.5
BUSINESSMAN	18	9.0
OTHERS	105	52.5
Total	200	100

Table-2.35.23 shows that most of the respondernts (38.5%) want to be government service holder, then (9%) want to be businessmen and (52.5%) respondents have other future plan in their life.

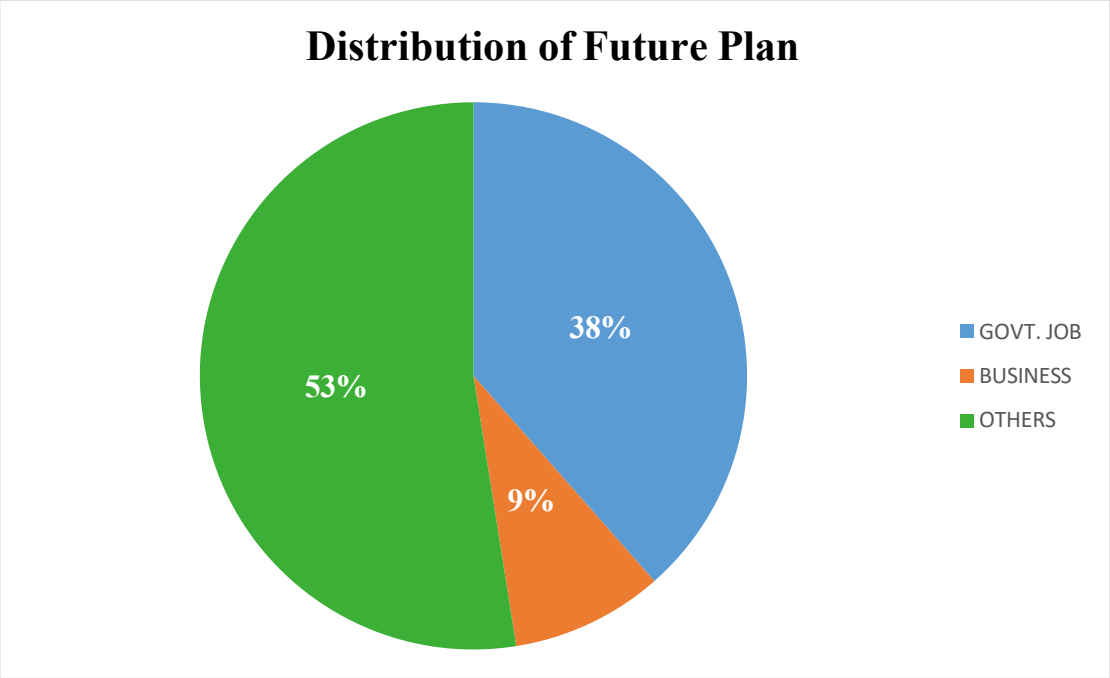

Figure-2.35.23: Pie-chart of future plan

Help to Achieve Goal

Table-2.36.24: Frequency distribution of Help to Achieve Goal:

Help to Achieve Goal	Frequency	Percent
No	60	30.0
Yes	140	70.0
Total	200	100

Table-2.36.24 shows that 70% respondents are agreed that involving in volunteer organization helps them to achieve their future goal and 30% are disagreed.

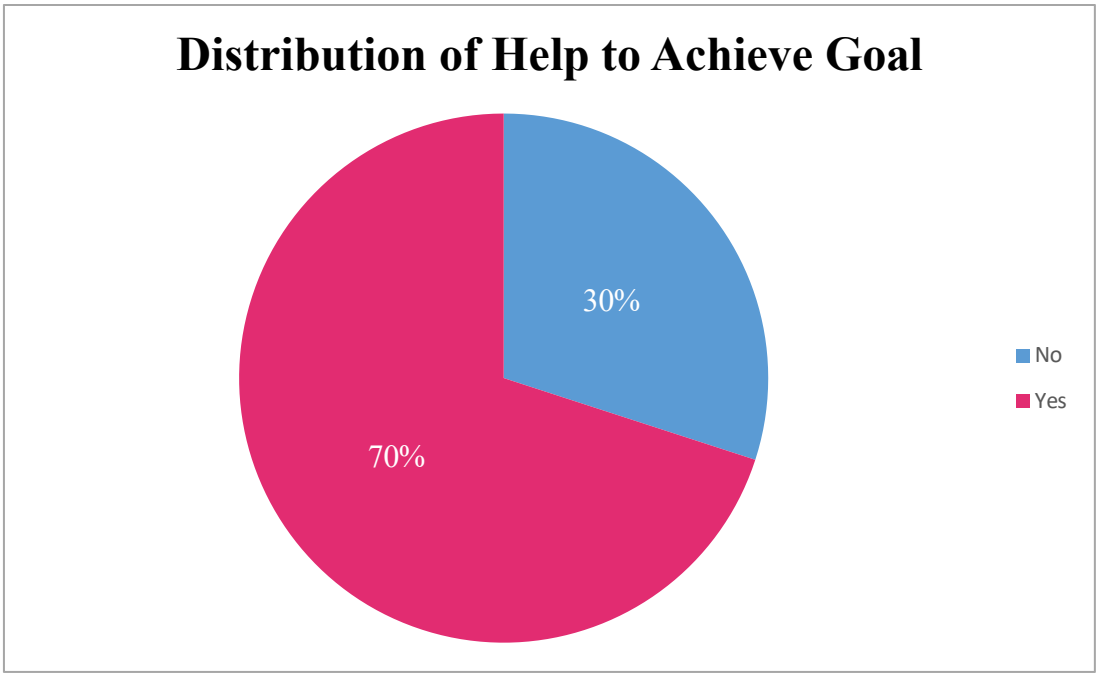

Figure-2.36.24: Pie-chart of help to achieve goal.

Academic Result

Table-2.37.25: Frequency distribution of Academic result:

Academic Result	Frequency	Percent
Less than or equal 3.5	115	57.5
3.50+	85	42.5
Total	200	100.0

Table-2.37.25 shows that most of the respondents (57.5%) academic results are below 3.5 and (42.5%) respondent's academic results are above 3.5.

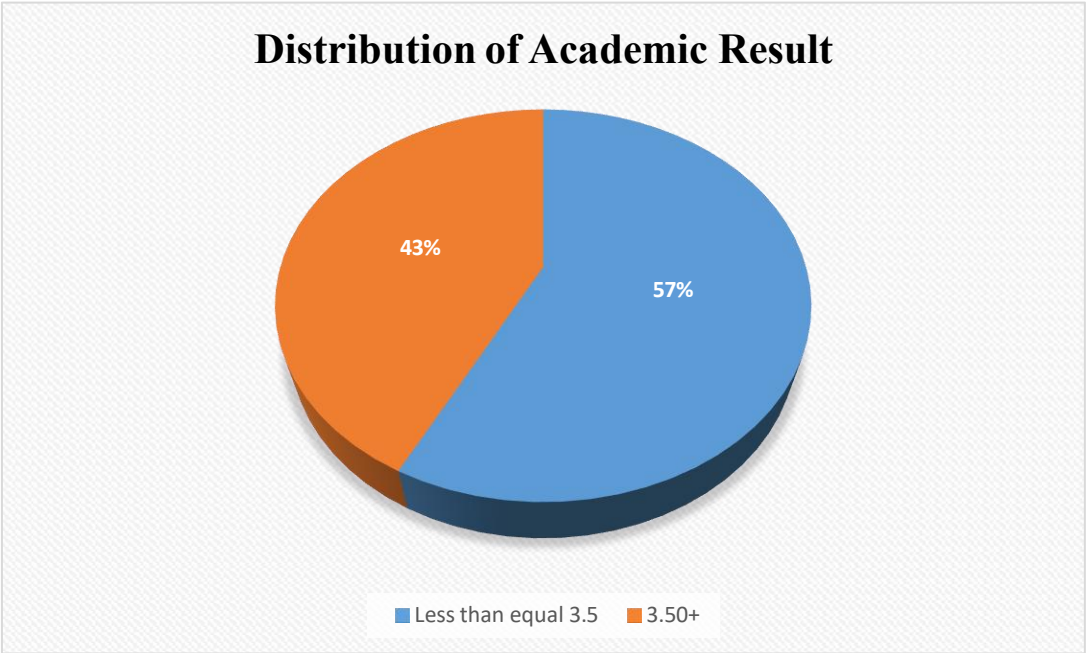

Figure-2.37.25: Pie-chart of Academic result

CHAPTER THREE

CONTINGENCY ANALYSIS

3.1 Introduction

Data analysis is the process of bringing order structure and meaning to the mass of collected data. There are many ways of analyzing data. Here, We use cross-table for our analysis.

From the cross tabulation alone, it is impossible to tell whether these difference are real or due to chance of variation. We can check the chi-square test to be sure. The chi-square test measures the discrepancy between the observed cell counts and what you would expect if the rows and columns were unrelated. Wef asymptotic significance (p-value) is less than α% where α is the level of significance, then null hypothesis may be rejected.

Chi-Square Test Formula:
$\chi^2 = \sum [(Oi - Ei)^2 / Ei] \sim \chi^2(r - 1)(c - 1)$
Explanation of Each Parameter:
χ^2: The chi-square test statistic. It measures how much the observed values deviate from the expected values under the null hypothesis.
∑: The summation symbol, meaning the sum is taken over all cells in the contingency table.
Oi: The observed frequency for the i-th cell in the table (actual data collected).
Ei: The expected frequency for the i-th cell, calculated assuming the row and column variables are independent. Ei = (row total × column total) / grand total.
r: Number of rows in the contingency table.
c: Number of columns in the contingency table.

$\chi^2(r - 1)(c - 1)$: The chi-square distribution with (r−1)(c−1) degrees of freedom. This is used to find the critical value or p-value for hypothesis testing

We need to take hypothesis to test the independency between them. Let us consider,

$_0$ = There is no relationship between them or there is insignificant relationship.

Vs. $_1$= Ho is not true.

Then we make a decision from the calculated χ^2 value or from corresponding p-value. If p-value ≤ α or $\chi^2_{cal.} \geq \chi^2_{tab.}$, reject H_0, otherwise accepted.

Table 3.2: Association between Spending time in organization and Academic Result:

H_0 : There is no association between spending time in organization and academic result.

vs

H_1: Null hypothesis is not true.

Spending_Time * Academic result Cross tabulation					
			Academic result		**Total**
			<=3.5	**3.5+**	
Spending_Time	Less than One hour	Count	17	23	40
		% within Spending_Time	42.5%	57.5%	100.0%
	One hour	Count	67	40	107
		% within Spending_Time	62.6%	37.4%	100.0%
	More than one our	Count	31	22	53
		% within Spending_Time	58.5%	41.5%	100.0%

Result of Chi-Square Test:

	Value	**df**	**Asymptotic Significance (2-sided)**
Pearson Chi-Square	4.851	2	.088

Table-3.2 shows that the Pearson chi-square statistic with 2 degrees of freedom is observed to be 4.851 which is asymptotically insignificant is 0.088 which is greater than the p-value 0.05. So we may accept our null hypothesis, which indicates that, there is no association between spending time in organization and academic result.

Table 3.3: Association between Having progress involving in volunteer organization and Academic result:

H_0 : There is no association between having progress and academic result.

vs

H_1: Null hypothesis is not true.

Academic result * Any_Progress_Happened Crosstabulation					
			Any_Progress_Happened		Total
			No	Yes	
Academic result	Less than or equal 3.5	Count	18	97	115
		% within Academic result	15.7%	84.3%	100.0%
	3.5+	Count	24	61	85
		% within Academic result	28.2%	71.8%	100.0%

Result of Chi-Square Test:

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	4.665[a]	1	.031

Table-3.3 shows that the Pearson chi-square statistic with 1 degrees of freedom is observed to be 4.665 which is asymptotically significant is 0.031 which is less than the p-value 0.05. So we may reject our null hypothesis, which indicates that, there is association between having progress and academic result.

CHAPTER FOUR
RESULTS AND DISCUSSION

The academic year distribution of the 200 respondents reveals a well-balanced representation across all levels of study at the university. Among them, **12.5% (25 students)** are enrolled in the **1st year**, just beginning their academic journey and likely new to volunteer activities. **18.5% (37 students)** are in the **2nd year**, where they begin to gain more exposure to university life and co-curricular engagements. The **3rd year** comprises **26.5% (53 students)**, reflecting a more experienced group with increasing involvement in volunteer work. The largest portion of respondents, **34.5% (69 students)**, are in their **4th year**, indicating a mature and academically focused group with high volunteer participation. Lastly, **8% (16 students)** belong to the **Master's level**, representing advanced students who have often continued their engagement in volunteerism from undergraduate levels and align their activities more directly with career aspirations.

The age group analysis reveals that a significant portion of the respondents **59.5%** are aged **between 21 to 23 years**, which corresponds closely with the academic standing of students in their **3rd and 4th years** of study. This age bracket reflects the typical age of students who are deeply engaged in both academic responsibilities and co-curricular activities such as volunteering. In addition to this dominant group, the study also includes **younger students from the 1st and 2nd years**, who are generally aged below 21 and are in the early stages of their university journey. On the other end, **older participants from the Master's level** are also represented, bringing a wider age diversity that contributes to a more comprehensive understanding of how volunteer involvement varies with academic maturity and life stage.

The religious background of the respondents indicates that a majority **85.5%** identify as **Muslim**, while the remaining **14.5%** belong to **other religious communities**. This distribution reflects the general demographic makeup of the student population at the university. In terms of gender,

89% of the participants are male, and only **11% are female**. Although the proportion of female respondents is relatively low, it is important to highlight that **female students are actively involved in meaningful and impactful volunteer roles**, particularly in **education-related initiatives** and **community outreach programs**. Their contributions, despite being numerically fewer, underscore the importance of inclusivity and the value of diverse participation in volunteerism.

30% of the respondents are from Rangpur Division, followed by others from Dhaka, Chattogram, Sylhet, Khulna, Barisal, Rajshahi, and Mymensingh. This shows a diverse representation from across Bangladesh.

The educational and occupational backgrounds of the respondents' parents provide insights into the socio-economic context of the student population. Among the respondents, **22% reported that their fathers had only primary level education**, while **31.5% of mothers** attained education up to the primary level. In terms of occupation, a significant portion **34.5% of fathers** are engaged in **farming**, and an overwhelming **86.5% of mothers** are **housewives**. Only a small fraction of fathers are employed in the **service or business sectors**, suggesting that the majority of students come from **rural or semi-urban areas with modest financial conditions**. Despite these limitations, it is noteworthy that these students are not only pursuing higher education but are also **actively involved in volunteer and co-curricular activities**, demonstrating strong personal motivation and a commitment to personal growth beyond academic requirements.

The family structure and income levels of the respondents offer a deeper understanding of their living conditions and socio-economic status. A significant majority **85.5%** of the students come from **nuclear families**, while the remaining **14.5%** belong to **joint family systems**. This distribution suggests a trend toward smaller, more independent family units among the student population. Regarding economic background, **36.5% of the students reported that their family's monthly income falls between BDT 10,001 and 20,000**, placing a large portion of them in the **middle to lower-income bracket**. Despite these financial constraints, the students

demonstrate a high level of involvement in volunteer work and co-curricular development, highlighting their resilience and strong desire to contribute to society while pursuing their education.

The study reveals that students are actively involved in a diverse range of volunteer organizations, reflecting their commitment to social responsibility and community service. Among these, **27% of the respondents are members of SCOUT**, which focuses on leadership, discipline, and public service. Many others are engaged with **NSTU Blood Brigade**, a well-known blood donation organization, as well as various **Social Welfare Clubs** and **university-based volunteering units** that operate across campuses to support numerous social causes. Notably, **41.5% of the students reported involvement in more than one organization**, highlighting a strong level of **multi-dimensional engagement**. This indicates not only their enthusiasm for volunteering but also their ability to manage time and responsibilities across different domains, which contributes significantly to their personal and professional development.

The analysis of time commitment to volunteer activities shows that **53.5% of students spend approximately one hour per day** engaged in various forms of community service. This consistent level of participation reflects a strong sense of discipline and structured **time management**, particularly impressive given their simultaneous academic responsibilities. The nature of their involvement is diverse and impactful, with **38% of students actively participating in blood donation drives**, contributing directly to lifesaving efforts. Additionally, students are involved in **admission assistance programs**, **child education initiatives**, **social awareness campaigns**, and **winter clothing distribution drives** for underprivileged communities. These activities not only demonstrate their dedication to social causes but also reveal how effectively students are balancing co-curricular engagement with their educational obligations.

The motivation behind students' involvement in volunteer activities is largely driven by **intrinsic interest**, with **44.5% of respondents indicating that they volunteer out of personal**

motivation rather than external pressure or requirement. This self-driven engagement reflects a deep sense of purpose and commitment to societal well-being. Among the various types of volunteer work, **social welfare activities emerged as the most preferred**, with **29.5% of students actively choosing this domain**, highlighting their desire to contribute meaningfully to the community. Significantly, **79% of the respondents reported that they have experienced progress either personally or professionally through their involvement in volunteer work**. A considerable portion, **36%**, specifically noted **communication skill development** as a direct and tangible outcome of their experiences, indicating that volunteerism not only supports community development but also enhances vital interpersonal and leadership skills among students.

70.5% of students were not involved in volunteering before entering higher education, meaning universities played a key role in introducing them to community service.

The study reveals that a significant portion of the student population **38.5%** has set **government service** as their primary career goal. This aspiration reflects a desire for stability, public impact, and meaningful civic contribution. Moreover, **70% of respondents stated that their involvement in volunteer organizations supports their journey toward achieving these goals**, indicating a **strong alignment between their co-curricular engagements and long-term life planning**. The structured environment, leadership opportunities, and community exposure provided by these volunteer platforms appear to play a crucial role in preparing students for competitive and service-oriented career paths, particularly within the public sector.

The academic performance of the respondents reveals that **42.5% of students have achieved a GPA above 3.50**, while **57.5% fall below this benchmark**. Although a slight majority have lower GPAs, the perception of volunteerism's impact on academics remains largely positive. In fact, **84.5% of the students reported that their participation in volunteer activities has not negatively affected their studies**. On the contrary, many believe that volunteering has enhanced their academic life by improving essential life skills. Students frequently mentioned **better time**

management, increased **leadership capabilities**, and a noticeable boost in **self-confidence** as outcomes of their involvement. These findings underscore the role of volunteerism not just as a co-curricular activity but as a valuable contributor to overall academic and personal development.

The **1st year students**, comprising **12.5% (25 students)** of the total respondents, are mostly new to university life and just beginning to adapt to the academic and social environment. Many of them have only recently joined volunteer organizations, and their level of involvement tends to be limited, with fewer participating in multiple organizations. At this early stage, the positive effects of volunteering—such as the development of communication or leadership skills—are not yet fully apparent. However, despite their newness to both university and volunteer work, the majority of these students reported **no negative impact** on their academic performance as a result of their participation.

The **2nd year students**, accounting for **18.5% (37 students)** of the respondents, demonstrate a growing involvement in volunteer organizations as they become more settled in university life. Compared to their first-year counterparts, they show a greater sense of responsibility and awareness, particularly in terms of **time management**. This group has started to engage in more **structured volunteer activities**, such as **blood donation drives**, **social welfare programs**, and other organized initiatives. While their academic results remain generally stable and unaffected by their volunteer participation, early signs of personal growth are evident. Many students in this group begin to show improvements in **communication skills**, **organizational abilities**, and **teamwork**, marking the beginning of their developmental journey through volunteering.

The **3rd year students**, making up **26.5% (53 students)** of the total sample, exhibit a noticeably higher level of engagement in volunteer activities. A substantial portion of this group is involved in **multiple volunteer organizations**, contributing significantly to the overall **41.5% multi-organization participation rate** observed in the study. Notably, this group reflects the most common volunteer time pattern, as **53.5% of all respondents spend about one hour daily** on

organizational work—a habit prevalent among third-year students. Their preferred activities include **blood donation**, with **38% participation**, and **social welfare initiatives**, with **29.5% involvement**, showing a clear dedication to community service. At this stage, the benefits of volunteering become more pronounced, with **36% of respondents reporting progress in soft skills**, especially in **communication, leadership, and organizational abilities**. This suggests that volunteer involvement is beginning to play a substantial role in shaping their personal and academic development.

The study surveyed **200 students** from various volunteer organizations to examine the **socio-economic, demographic, and academic characteristics** of respondents. The **4th-year students**, **50%** are from the **science faculty**, and **34.5%(69 students)** are the largest group, known for their maturity and intrinsic motivation toward volunteering. Most students (**59.5%**) are aged **21–23**, **85.5%** are **Muslim**, and **89%** are **male**. A significant number come from modest backgrounds: **34.5% of fathers** are farmers, **86.5% of mothers** are housewives, and **36.5%** of families earn between **BDT 10,001–20,000** monthly.In terms of involvement, **27%** volunteer in **SCOUT**, **41.5%** are part of multiple organizations, and **53.5%** dedicate **1 hour daily** to volunteer work. Popular activities include **blood donation (38%)** and **social welfare (29.5%)**. Despite their involvement, **84.5%** reported **no negative academic impact**, and **79%** said volunteering helped them progress personally or professionally. Notably, **44.5%** volunteer from personal interest, and **36%** noted improved **communication skills**. While **57.5%** of students have a GPA below 3.50, the majority find that volunteering **supports their academic and career goals**, with **70%** saying it aids their **future plans**, especially those aiming for **government service (38.5%)**.

The **Master's students**, comprising **8% (16 students)** of the total respondents, form the smallest yet one of the most experienced groups in this study. Many of these students have likely continued their **volunteering roles from their undergraduate years**, building on a strong foundation of co-curricular involvement. At this advanced academic stage, their focus

tends to shift toward **career alignment**, and this is clearly reflected in the findings—**70% of them reported that their volunteer experiences directly support their long-term goals**. These students often exhibit the most refined balance between academics and volunteerism, effectively integrating both aspects into their personal and professional development. Their journey illustrates how sustained volunteer involvement can complement academic growth and enhance real-world readiness.

The cross-tabulation analysis of this study yielded **two key statistical conclusions** regarding the relationship between volunteer involvement and academic performance. The first comparison examined the **amount of time spent volunteering and its impact on academic results**, resulting in a **chi-square value of 4.851** with a **p-value of 0.088**. Since this p-value is greater than the 0.05 threshold, the result is **not statistically significant**, indicating that there is **no strong evidence to suggest that increased time spent in volunteer organizations negatively affects academic performance**. In contrast, the second analysis explored the **relationship between students' perceived progress such as improvements in leadership, communication, and other soft skills and their academic outcomes**. This test produced a **chi-square value of 4.665** with a **p-value of 0.031**, which is statistically significant. This finding reveals a **meaningful association between students who report personal and skill development through volunteering and those who achieve better academic performance**, suggesting that such experiences contribute positively to their educational success.

Students across **all academic years** benefit from volunteering in terms of personal development and soft skill acquisition. Despite differences in experience and engagement level, the **academic performance is not negatively affected** by volunteering time.In fact, students who claim to have made **life progress** through volunteer activities show a **statistically significant** tendency to have better academic results.Therefore, volunteering is not just harmless to academic outcomes ,it is actually **developmentally beneficial**, especially as students mature through each academic year.

CHAPTER FIVE
CONCLUSION

This study highlights the significant impact of student involvement in volunteer organizations on both academic success and personal development. Among the 200 respondents from various faculties and academic years, a consistent pattern emerged: volunteering fosters growth in communication, leadership, and time management skills, all of which contribute positively to students' academic and career trajectories.

Most students entered university with no prior volunteer experience but became involved through personal interest during higher education. Notably, **44.5%** of respondents volunteer out of intrinsic motivation, and **79%** reported noticeable personal or professional progress through these activities. Volunteering avenues such as **blood donation, child education, winter clothing drives, and admission support** provide meaningful exposure that enhances empathy, discipline, and organizational abilities.

Crucially, **84.5% of students confirmed that volunteering does not negatively affect their academic performance**. A chi-square analysis supports this, showing no significant relationship between time spent volunteering and GPA ($p = 0.088$). However, a statistically significant association ($p = 0.031$) exists between self-reported progress and higher academic achievement, affirming that students who gain leadership and communication skills from volunteering often excel academically.

These benefits are observed across all academic years and faculties ,students from Science, Business, Education, and Social Science report similar outcomes. **Fourth-year students**, representing the largest group, exhibit the highest maturity in balancing volunteering with academics, further proving that these activities can coexist effectively.

In summary, student volunteerism is not a distraction ,it is a **valuable educational supplement**. It develops well-rounded individuals equipped not only with academic knowledge but also real-world skills. As such, educational institutions should encourage structured volunteering opportunities to prepare students for both **professional success** and **social responsibility**, nurturing a generation that is academically competent and civically engaged.

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